ESSENTIAL FEATURES OF INTELSAT: APPLICATIONS FOR THE FUTURE

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Introduction

International cooperative activities in outer space are increasing and will require a variety of institutional forms, mechanisms and approaches. This article will focus on some concepts concerning the ways in which these activities can be organized. These concepts are drawn from the experience of one particular international organization, the International Telecommunications Satellite Organization (INTELSAT),[1] dealing with one particular set of space-related activities: space communications. This experience, however, may have some broader applicability not only to space-related activities but also to other areas of international cooperation as well.

It should be emphasized that INTELSAT is not a model. It is not possible to take one organization developed in one context and at a particular time and simply apply it in another context and at another time. Morever, INTELSAT deals with satellite communications, which is only one small part of the range of outer space activities. But the particular institutional features and mechanisms of INTELSAT are not limited to or only reflective of problems concerning satellite communications. Rather, they can be regarded as responses or solutions to particular sets of problems or needs.

This article addresses five specific problem areas: *first,* the question of timing the institutional arrangements; *second,* the question of combining political and technical aspects; *third,* the question of management; *fourth,* the interrelated questions of finance and control; and *fifth* the matter of technology transfer.

1. The Problem of Timing the Institutional Arrangements

It goes without saying that it is not easy to judge the right time to establish an international institution and its set of arrangements. In areas with a heavy technological component, there is the risk of being premature and acting before the technology, or perhaps more accurately, the application of the technology, has sufficiently jelled to permit a realistic assessment of the most appropriate institutional arrangements. Sometimes, even if the technology is defined and its application is clear, it may not be possible, for political or other reasons, to reach agreement on institutional arrangements. In short, institutional arrangements do not exist in a vacuum.

It also may not be wise to place too much reliance on the existence of an amendment process later to revise institutional arrangements in the light of experience. Many international agreements, particularly those establishing institutional arrangements, are a bundle of interrelated compromises adopted as a single package

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[1]Agreement Relating to the International Telecommunications Satellite Organization, INTELSAT, August 20, 1971, [1971] 23 U.S.T. 3813, T.I.A.S. 7532.

after a long struggle. It may be very hard to start amending one aspect without having a "ripple effect" on other interrelated aspects. Moreover, once a precedent is established to commence the process of amending an agreement, the door may be opened to frequent amendments or attempts at amendments on a wide range of matters. Consequently, there may be a strong reluctance to commence using the amendment process for adjusting the institutional arrangements to changing conditions, even if specific changes can be amply justified.

Concerning timing, the choice does not have to be between all or nothing—between *either* establishing a complete set of international arrangements *or* deferring any action until a later date. This *either/or* choice is a fallacy. In this respect, one aspect of the INTELSAT arrangements—involving a phased transition in the establishment of the Organization—may have some general applicability.

INTELSAT was established on an interim basis in 1964 by an "Agreement Establishing Interim Arrangements for a Global Commercial Communications Satellite Consortium,"[2] which clearly contemplated that it would lead to a permanent or definitive form of organization. A time period was specified in the Agreement for a study to be conducted by the international governing body, called the Interim Committee, and submitted to all members. The report was to consider: "among other things, whether the interim arrangements should be continued on a permanent basis or whether a permanent international organization with a General Conference and an international administrative and technical staff should be established."[3]

The 1964 Agreement provided for the convening of an international conference to consider the report and set a target date of 1970 for the entry into force of definitive arrangements adopted by the Conference. However, no fixed deadline was set for the end of the interim arrangements and for the transition to permanent arrangements; in the absence of agreement, the interim arrangements as a formal matter could have continued indefinitely.

In 1964, when the interim arrangements were established, it was premature, both technically and politically, to establish a permanent international organization with a sufficiently large degree of authority to manage fully a complex activity such as the global communications satellite system. In that early stage of space activities, the technology and expertise resided largely in one country, the United States. At the same time, the nature of the activity—establishing paths of communications between nations—required some form of international cooperation. Thus, it was not too early to set up some international framework and to make provision for a later transition to a more permanent arrangement.

The international negotiations to establish these permanent arrangements, which stretched over more than a two-year period from 1969 to 1971, were far harder to accomplish than setting up interim arrangements of course, because the arrangements were to be permanent. Yet here, too, the issue that proved to be the most difficult one to resolve concerned the shift from management of the system by a national entity to management by the international organization itself. The *either/or* dilemma referred to

[2]Agreement Establishing Interim Arrangements for a Global Communications Satellite System, [1964] 15(2) U.S.T. 1705.

[3]*Id.*

above was avoided and a six-year transition scheme was worked out (as described more fully below).

The applicability of this type of phased-timing arrangement to other areas depends on many factors, such as the technology involved, the political situation, and the prior history of cooperation. For example, in the case of the international organization created to own and operate a communications satellite system to provide maritime services, the situation was quite different: more than a decade had passed in which the technology had matured and its application had been defined, and experience had been gained with the workability of institutional arrangements in the form of INTELSAT. Accordingly, there was not the same need to proceed more slowly with a phased transition into an international organization. In other activities which are yet to emerge, such as space platforms, the situation undoubtedly will be different, calling for different arrangements.

2. The Question of Combining Political and Technical Elements

There are many international cooperative activities in technical areas, such as health, aviation, food standards, communications, energy, etc. Much of the work at the national level in connection with these activities is done primarily by the technical government ministries and the departments concerned rather than by the foreign ministries. This work includes development of positions, representation at international meetings and subsequent implementation of the international standards, regulations or recommendations.

It is a fallacy to assume that in practice, governments are monoliths, speaking with one unified voice. Very often, the technical experts and representatives to international conferences may have more in common with each other than each may have with their respective foreign ministries. In view of the common technical background and education, shared professional concerns and goals are understandable and not at all undesirable. At the same time, however, coordination of the technical and political elements is obviously necessary at the national level as reflected in the various governmental ministries. The question then becomes how appropriately to combine both of these elements in an institutional framework concerned with international cooperative activities. INTELSAT provides one example of how this question has been answered.

INTELSAT is governed by two international agreements: the Agreement setting forth the basic provisions and principles and structure of the organization, signed by governments through their foreign ministries, and an Operating Agreement setting forth more detailed financial and technical provisions and signed by the governments or their designated telecommunications entities. In most cases these Signatories are the ministries of communications of the countries concerned, but in some cases they are quasi-public or private entities under varying degrees of governmental control or supervision. (The most prominant example in this respect is the Signatory representing the government of the United States in the Organization—the Communications Satellite Corporation, a private corporation established by federal legislation to represent the United States in the international arrangements for the global communications satellite system.)

INTELSAT's structure contains three representative organs which reflect this distinction between political and technical representation. First, the Assembly of Parties, which meets every two years, is composed of all States party to the Agreement, and is primarily concerned with "those aspects of INTELSAT which are primarily of interest to the Parties as sovereign States." Decisions are taken on a one-country, one-vote basis, and the principal representation is provided by the foreign ministries.

Second, the Meeting of Signatories, which meets annually, is composed of all the Signatories to the Operating Agreement, and is primarily concerned with financial, technical, and program matters of a general nature. It is empowered to take certain specified decisions of a general nature in these areas, such as raising the capital ceiling and establishing general principles for utilization charges. Decisions are taken on a one-country, one-vote basis, and representation is provided by the Signatories, for the most part, communications ministries; thus, it is more technically and operationally oriented than the Assembly of Parties.

Third, the Board of Governors, which meets at least four times a year, is the organ with the basic responsibility to take decisions covering the design, development, establishment, operation and maintenance of the space segment. It is composed of Governors representing Signatories with a certain minimum investment share, singly or in combination, and, with certain limitations, Governors representing at least five or more Signatories. Decisions are taken by weighted vote, each Governor casting a vote equivalent to his Signatory's investment share. The representatives of the Signatories to the Board of Governors are officials concerned, in their home countries, with the operation and management of the satellite system, and generally have a management, technical, or financial background. The Board, in turn, is assisted by several advisory committees covering technical, financial, and planning matters, and these groups are comprised of specialists from their respective Signatory organizations.

This arrangement works well in practice. Decisions are considered and taken by technically oriented representatives who will have the responsibility, within their home countries, for implementing these decisions (e.g., the construction of new earth stations), and who have a direct and practical common interest in the continued viability of the operational system. At the same time, recognizing that many decisions of a "technical" nature have interwoven political and foreign policy aspects, there is, of course, often consultation and coordination on particular issues by the Governors and representatives to the Meeting of Signatories with their respective foreign ministries.

The arrangement works because of the large degree of common interest among the member states and because the subject matter is not one in which controversial issues of a political nature predominate. Whether this type of arrangement would work as well in other areas would depend on such factors as the nature of the membership, the political issues involved, and the extent to which a shared common interest existed.

3. The Question of Management

The question here is management on an international level of an operational system. It is not a question of an international secretariat: almost all international organizations have such secretariats, but such organizations are primarily concerned with regulatory or coordinating functions. With the exception of the World Bank and its associated institutions, there are almost no intergovernmental organizations with a

direct operational responsibility. Thus, INTELSAT's experience as an international organization with the direct responsibility of owning and managing a global communications satellite system may be instructive and of some applicability in other situations.

As discussed earlier in connection with the question of timing, it is important to avoid an *either/or* dichotomy: either the international organization is entrusted with management functions at one time or not at all, and either it will be entrusted with all management functions or with none. As to both of these choices, the INTELSAT experience indicates that there are a wide range of alternatives which avoid this dichotomy. The issues are to what extent, how fast, and under what terms should effective management control be shifted from the national to the international level? These are, of course, as much policy issues as technical or management issues.

Various transition schemes are possible to time the introduction of international management arrangements. A principal feature of the permanent INTELSAT Agreements was the provision for a phased shift from national to international management over a seven-year period.

In the first phase, which extended from 1973, when the organization was first established, to the end of 1976, the international staff was headed by a Secretary General and a relatively small support staff. However, the technical and operational management of the system was entrusted to the Communications Satellite Corporation, which was the U.S. Signatory to the Operating Agreement and had served as Manager of the global system under the interim arrangements in force from 1964 to 1973. The INTELSAT Secretary General had no direct authority to supervise this contractor, which reported directly to the Board of Governors, but had a monitoring role to keep the Board informed about its activities.

The second phase, from 1976 to 1979, saw a further shift in the direction of international management. The Secretary General became the Director General with responsibility for all management services and for supervision of the performance of the management services contractor. While the monitoring and reporting relationships in this interim period were complex and would not have been devised by an efficiency expert, they did reflect the realities of the situation.

The third phase, which commenced in 1979, in a sense marked the end of the period of transition to international management. The management contract with a national entity had ended, but the Agreements provided some criteria to help determine whether all management functions would be conducted in-house or not. Article XI(c) (ii) provided:

> The Director General, on behalf of INTELSAT, shall contract out, to one or more competent entities, technical and operational functions to the maximum extent practicable with due regard to cost and consistent with competence, effectiveness and efficiency. Such entities may be of various nationalities or may be an international corporation owned and controlled by INTELSAT. Such contracts shall be negotiated, executed and administered by the Director General.

A significant portion of the functions previously contracted out were now assumed by the Organization itself, with a consequent major increase in staff, including the *en masse* transfer from the contractor of the technical personnel who had previously performed these functions. As part of this process, an expanded and revised organizational structure for the Executive Organ was also established.

In addition, in applying the criteria specified in the Agreement (and quoted above), INTELSAT concluded two technical services contracts with the same entity that had previously provided management services to provide assistance in certain defined areas for periods of six and four years. An additional contract was also concluded for specified laboratory services for a three-year period. These contracts differed from the preceding management service contracts essentially in that the scope and discretion of the contractor was greatly reduced and its role was confined to the performance of specified tasks.

As these contracts come up for review in the next few years, judgments will have to be made, in light of the above-quoted, mandated criteria, as to which specific functions should be brought in-house, which should continue to be contracted out, to whom, and under what arrangements.

Quite obviously, the management arrangements just described and their transitional aspects reflect a particular situation and cannot automatically be applied in other situations. For one thing, the international organization inherited an ongoing operational system which was already under the effective management of a national entity and upon which many countries relied for their telecommunications. Time was needed to build up an international staff and to gain the requisite experience in this highly specialized activity.

In other areas, where an international organization could be put in place *before* an operational system was established or complex activities were commenced, it might not be necessary to have so elaborate a transition scheme. Moreover, in certain areas, depending on the state of the technology and its distribution, the question of the extent to which certain technical and management functions should be contracted out to national or other international entities could be resolved differently. What seems clear, however, is that if there is to be a division of the activities in question between the international organization, on the one hand, and other entities, whether national or international, on the other hand, the key to success is close cooperation and effective working relationships, in view of the interrelated nature of these activities.

4. Finance and Control

The problems are: how is power and authority to be distributed among the member states of an international organization? What criteria should apply?

Since the financial contribution and status of the members are different, one cannot reasonably have a one-country, one-vote rule for those decisions having major financial consequences. (At the same time, one must recognize the basic principle of nondiscriminatory access by all members to the benefits and technology resulting from the Organization's activities.)

It is easier, of course, to state the problem than to devise a workable solution. As always, the particular solution will depend on the specific situation and context. But here again, INTELSAT's experience may be instructive.

INTELSAT's basic philosophy is that each member's financial interest in the Organization—its investment share—is strictly proportional to its use of the system, as determined on a annual basis. If member X's use of the system, as measured in revenues paid in a given year, is, let us say, 9%, then its investment share is 9%, and it is required to contribute 9% of the costs incurred by the Organization in carrying out its

activities. The corollary concept is that each member's vote in the Board of Governors, the key organ that takes the key decisions relating to the establishment of the system, including those having financial consequences, is based on its investment share. As a member's proportionate use of the system will fluctuate over the years, so its investment share in the Organization and its voting power on the Board will also fluctuate. (There are some exceptions to this basic principle, as applied to voting, to reflect the interests in representation on the Board of those countries with minimum financial status in the Organization, and in terms of the majorities necessary to adopt decisions.)

In practice, decisions are rarely taken on the basis of votes, but are normally arrived at by consensus after a process of consideration and consultation over a period of several meetings. This is especially true for significant decisions with major financial consequences.

5. Transfer of Technology

In the course of international cooperative work, particularly in high technology areas such as space activities, advanced technology in the form of information and data will be developed. The questions then are: how are the members of the Organization, which financed the activity that produced the technology able to share in its fruits? What rights should they have, what restrictions should be imposed, and how is the information to be transferred to them?

INTELSAT has had some fairly elaborate experience in this area. Under its interim arrangements, during the period 1964-1973, the basic principle was that INTELSAT itself would own the technical data and inventions that arose from work it financed, such as contracts for the manufacture of spacecraft and research and development contracts. In light of experience, this policy was changed in the permanent arrangements, which entered into force in 1973, so that INTELSAT obtained only a non-exclusive, royalty-free license to use the technology, with the developer retaining ownership rights.

Under both systems, an elaborate procedure was developed to identify the technical information and data and then to make arrangements for its transfer to those members wishing to utilize it and to safeguard, from a proprietary standpoint, INTELSAT's rights in such data. For example, determinations need to be made as to whether a particular invention should be the subject of a patent application, and if so, in which countries and which respective Signatories should be notified.

The arrangements, though complex, have worked reasonably well in practice in terms of an effective transfer of the technology to members wishing to utilize it. One area where practical solutions still need to be found is in the application of national laws concerning dissemination of data. The applicability of this type of arrangement to other areas would depend, among other factors, on the nature of the technology in question and the degree of its demand by member states. In INTELSAT's case, while INTELSAT's members benefited from the application of this technology to the global system, as a general matter, they did not have the capability or need to utilize it directly themselves, apart from the Organization. In other situations where such a widespread need might exist, the arrangements might well need to be different.